THE LONGEVITY EQUATION

Uncovering the Hidden Forces of Human Lifespan

By Liu Tingsen, Lin Xiaoya, Wu Yiqian

1. Overview & The Original Idea

1.1 Introduction

We have always been told that wealth equals health. It is a universal assumption that economic development, often measured through GDP per capita, inevitably improves living standards and human longevity. But what happens when the data disagrees? What happens when you look closely at some of the world's wealthiest nations and realize their citizens are dying a decade earlier than they should be?

"The Longevity Equation" was born from this exact curiosity. We wanted to move beyond the traditional, encyclopedia-style dashboards offered by platforms like Gapminder. Our goal was to create a guided, interactive data journalism piece that investigates the intersection of national wealth, gender demographics, and chronic lifestyle diseases.

1.2 The Dataset

To build a comprehensive narrative, we combined three publicly available, authoritative datasets spanning from 2000 to 2021:

- Life Expectancy at Birth from the World Health Organization (WHO).
- GDP per Capita from the World Bank.
- NCD Mortality Rate (Mortality from Non-Communicable Diseases like cardiovascular disease, cancer, and diabetes) from the World Bank.

By merging these datasets, we created a unified foundation of over 12,000 records. This allowed us to explore not just how long people live, but why they might be dying prematurely.

1.3 Target Audience

Our target audience includes students of global development, public health enthusiasts, and the general public. We designed the experience to be intuitive, ensuring that a user with no statistical background can easily grasp complex socio-economic concepts.

2. The Journey: From Concept to Reality

2.1 Initial Brainstorming

When we first conceptualized this project during Milestone 1, we asked three central questions:

1. How strongly is GDP associated with life expectancy?
2. What is the persistent gap between male and female longevity?
3. Does higher income automatically imply lower mortality from preventable diseases?

Initially, we imagined a highly technical dashboard. We envisioned multi-layered charts and complex cross-filtering systems. However, as we progressed through the design phase, we realized that presenting the user with a wall of charts is intimidating.

2.2 The Scrollytelling Pivot

We decided to adopt a "Scrollytelling" architecture. Inspired by data journalism outlets like The Pudding, we built a custom paginated navigation system. We wanted to control the narrative flow, guiding the user step-by-step through global demographic trends before landing on our primary finding: The Wealth Paradox.

Instead of letting users drown in data, we introduce concepts sequentially: first Biology (Gender), then Geography (Map & Rankings), then Economics (The Paradox), and finally, a Personal Reflection.

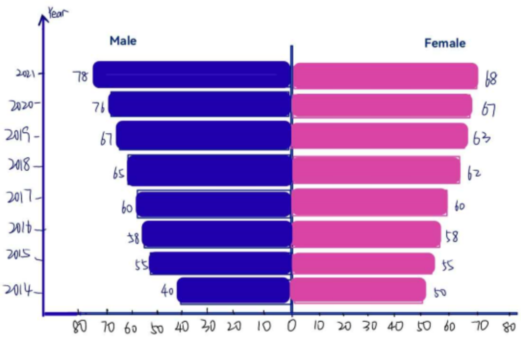
3. Evolution of Visualizations

3.1 The Gender Divide: From Concept to Butterfly Chart

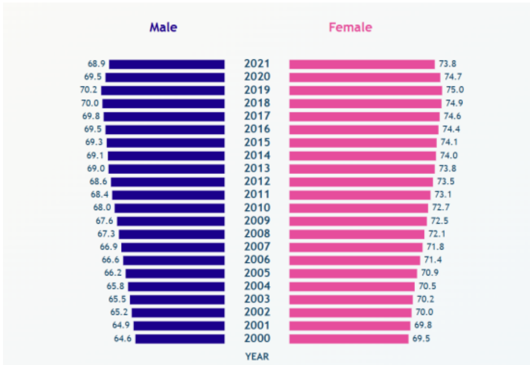
The Original Idea

We knew we wanted to show the difference between male and female life expectancy. Initially, we considered standard grouped bar charts or simple line graphs showing trends over time.

The Evolution



Sketch of Gender Chart



Final D3 Butterfly Chart Screenshot

We ultimately decided on a Butterfly Chart (also known as a Tornado Chart). By having the male bars extend to the left and female bars extend to the right, the persistent global survival gap becomes immediately visually striking. During development, we ensured the most recent year (2021) was placed at the top, following the reverse-chronological reading habit of digital journalism.

3.2 Global Presence: Choropleth & Racing Bars

The Original Idea

We wanted to give users a macro-level view of global health progression over the last two decades.

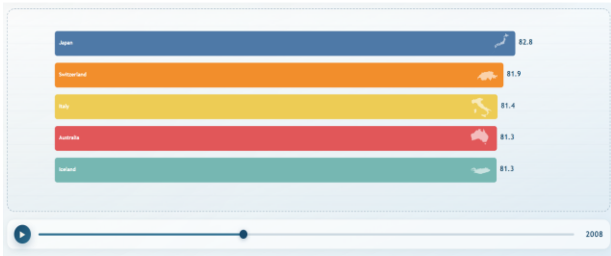
The Evolution

(Insert Sketch of Map/Top 5 -> Insert Final D3 Map & Bar Chart Screenshot)

Sketch of Map

Final D3 Map Screenshot

Sketch of Map/Top 5



Final D3 Bar Chart Screenshot

We implemented an interactive Choropleth Map linked to a manual timeline slider. As the user scrubs through the years, the world transitions from darker to lighter colors, illustrating global health improvements. To add a dynamic, competitive element, we paired this with a Racing Bar Chart. As the years tick by, the top 5 countries automatically reorder themselves using smooth D3.js transitions, turning dry historical data into an engaging visual race.

3.3 The Wealth Paradox: Uncovering the Anomaly

The Original Idea

We wanted to show that GDP and life expectancy are correlated, but not perfectly. Our analysis featured scatter plot and bar chart with relevant statistics.

The Evolution

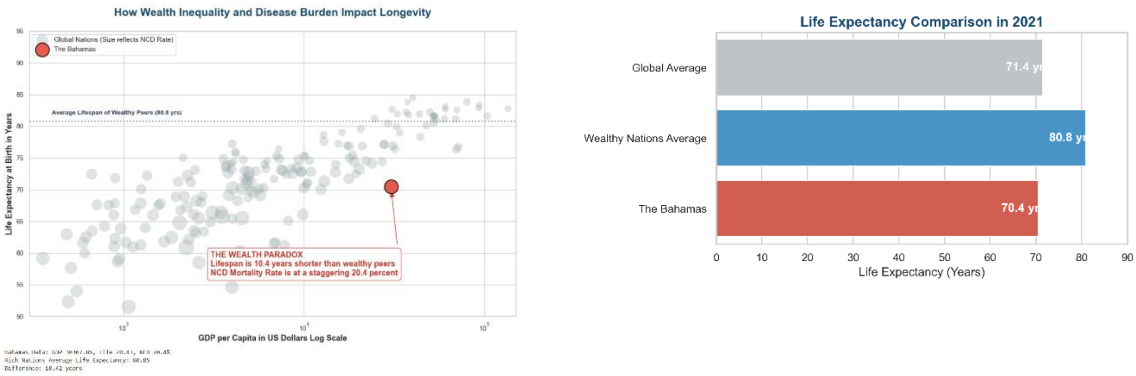


Figure 1. The Wealth Paradox Analysis (The Bahamas)

This became the climax of our data story. We isolated The Bahamas as our primary case study which is a nation with a GDP per capita of over 30,000 dollars, but a lifespan nearly 10 years shorter than its wealthy peers.

To visualize this, we used a logarithmic Scatter Plot. But we added a crucial third dimension: Bubble Size. We encoded the Non-Communicable Disease (NCD) mortality rate into the size of the scatter points. Suddenly, the paradox made sense visually. The Bahamas sits far below the global trendline, and its bubble is massive compared to its neighbors. This proved our hypothesis: raw wealth cannot buy time if a nation suffers from extreme inequality and high chronic disease rates.

4. A Major Design Pivot: Redesigning the Conclusion

4.1 Abandoning the Radar Chart

One of our most significant design decisions occurred between Milestone 2 and Milestone 3.

The Original Plan

Our initial sketches for the final section featured a "Multidimensional Radar Chart." We wanted users to select a country and see a polygon mapping its Wealth, Overall Lifespan, Male Lifespan, Female Lifespan, GDP per capita and NCD Rate.

The Problem (A Flawed Concept)

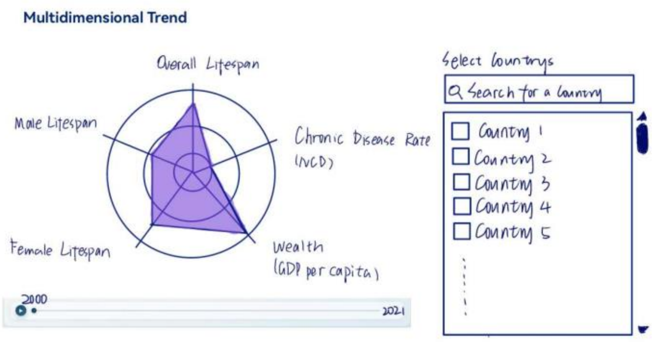


Figure 2. Original Sketch of the Radar Chart

As we began analyzing the data to code the D3 prototype, we realized this design violated several core data visualization principles:

1. **Mismatched Units:** Plotting GDP (which scales into tens of thousands) on the same geometric plane as Life Expectancy (which caps at around 85) causes severe distortion. Even with mathematical normalization, the resulting polygons felt abstract and disconnected from the human experience.
2. **Logical Redundancy:** We realized that plotting "Overall Lifespan" on the same axes as "Male Lifespan" and "Female Lifespan" makes no statistical sense. The overall metric is simply an average of the two genders, meaning the axes would overlap and artificially skew the shape of the polygon.

Realizing that this visualization would confuse rather than clarify, we made the bold decision to scrap the Radar Chart entirely.

4.2 The Solution: "Where Do You Stand?"

Instead, we asked ourselves: What does the user actually care about at the end of this story? The answer is Themselves!

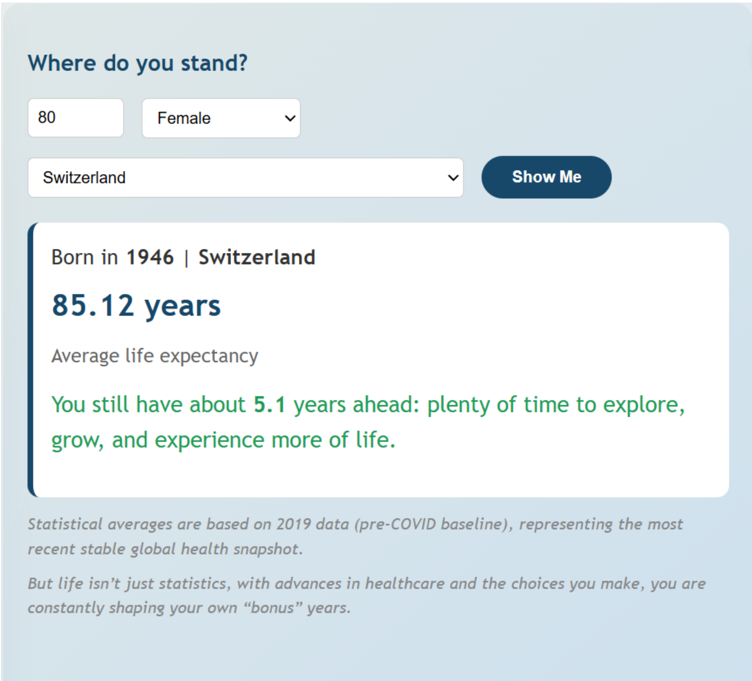


Figure 3. Final Personal Calculator Input Form Screenshot

We pivoted to a Personal Longevity Calculator. We built an interactive interface where users input their age, gender, and country. Using our pre-COVID (2019) baseline data, the tool calculates their statistical life expectancy, compares them to the global average, and reveals their "bonus years" based on their demographic footprint. This transformed a cold, statistical conclusion into a highly engaging, personal reflection.

5. Technical Challenges & Solutions

Building a seamless, interactive experience using purely native technologies presented several hurdles.

5.1 Merging Incompatible Datasets (EDA)

Our raw data came from two different massive institutions (WHO and World Bank). Country naming conventions varied wildly (e.g., "United States" vs. "USA" vs. "United States of America").

Solution: During our Python EDA phase, we had to heavily clean the data, standardize country codes (ISO-3), and perform inner joins. We also applied logarithmic transformations to GDP data to reduce skewness before exporting the final clean JSON files for our frontend.

5.2 Transition Animations

We wanted the user experience to feel "alive." However, managing the update pattern in D3 (Enter, Update, Exit) for the Racing Bar Chart and the Scatter Plot was complex.

Solution: We utilized D3's .transition(), .duration(), and .ease() functions to interpolate data points. This ensured that when a user moved the timeline slider, the bars and bubbles didn't just snap to new positions, but smoothly glided, helping the user track data movement visually.

6. Visual Language & Architecture

6.1 Color Palette & Typography

We approached the design of this website with the mindset of a digital magazine. We chose a palette that balances professionalism with emotional impact:

- Deep Navy Blue (#17486a) : Used for primary text and headings to establish trust and academic rigor.
- Muted Grays (#95a5a6) : Used for background data points (like the global nations in the scatter plot) to reduce visual noise.
- Crimson Red (#e74c3c) : Used sparingly but strategically as our "Accent" color. We used this red exclusively to highlight anomalies like The Bahamas, forcing the user's eye directly to the most critical data points.

We opted for clean, sans-serif fonts to ensure readability across all devices. We used varying font weights (Light, Regular, Bold) to establish a clear visual hierarchy, ensuring that the narrative text guided the user effortlessly toward the interactive D3 elements.

6.2 Frontend Architecture

By utilizing Vanilla HTML, CSS, and JS without heavy frameworks like React, we ensured our site remained incredibly fast and lightweight. The DOM manipulation is handled natively, allowing the D3.js engine to perform at maximum efficiency without rendering conflicts.

7. Peer Assessment & Conclusion

7.1 Team Contributions

This project was a collaborative effort, with each team member bringing distinct expertise to the table:

- **Lin Xiaoya (423134):** Xiaoya conducted the extensive data preprocessing and Exploratory Data Analysis (EDA) in Python. She authored the Milestone 2 report, the final Milestone 3 Process Book, Milestone 3 Screencast, readme.md on GitHub repository, and all the narrative text found on the website.
- **Liu Tingsen (422014):** Tingsen developed the custom HTML and Vanilla JS logic that powers the scrollytelling navigation. He was also responsible for drafting the initial Milestone 1 proposals.
- **Wu Yiqian (423147):** Yiqian conducted the related work research and took charge of coding the core D3.js visualizations. She was responsible for adjusting the final structure and integrating the D3 widgets into the responsive containers.

(Note: While tasks were divided for efficiency, all design decisions, architectural pivots, and narrative directions were debated and agreed upon collectively.)

7.2 Conclusion & Future Work

"The Longevity Equation" successfully transforms raw, intimidating health data into an accessible, human-centric story. We proved that data visualization is not just about drawing charts, it is about guiding a user toward a revelation.

If we had an additional semester to expand this project, we would look into incorporating regional, sub-national data (e.g., state-by-state data in the US or provinces in China) to uncover inequalities within borders, rather than just between them. We would also explore adding an API connection to pull live demographic data automatically.

Ultimately, we are incredibly proud of the interactive journey we have built.